

```

7
8  /**
9   *
10  * @author Piotr
11  */
12  public class Group {
13
14      private String name;
15      private int cardinality;
16      private Student[] students;
17      private Test[] tests = new Test[20];
18      private int testCounter;
19      private String[] sNames;
20
21      @Override
22      public String toString() {
23          return "Group{" + "name=" + name + ", cardinality=" + cardinality + '}';
24      }
25
26      public Group(String name, int cardinality) {
27          this.name = name;
28          this.cardinality = cardinality;
29
30          students = new Student[cardinality];
31          for (int i = 0; i < cardinality; i++) {
32              students[i] = new Student();
33          }
34      }
35
36      public Group() {
37          name = "";
38          cardinality = 0;
39          testCounter = 0;
40      }
41
42      public Student getAtt(int i) {
43          return students[i];
44      }
45
46      public String[] getsNames() {
47          return sNames;
48      }
49
50      public void setsNames(String[] sNames) {
51          this.sNames = sNames;
52      }
53
54      public void setName(String name) {
55          this.name = name;
56      }
60
61  }

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```
61
62  public void setStudents(Student[] students) {
63      this.students = students;
64  }
65
66  public void setTests(Test[] tests) {
67      this.tests = tests;
68  }
69
70  public String getName() {
71      return name;
72  }
73
74  public int getCardinality() {
75      return cardinality;
76  }
77
78  public Student[] getStudents() {
79      return students;
80  }
81
82  public Test[] getTests() {
83      return tests;
84  }
85
86  public int getTestCounter() {
87      return testCounter;
88  }
89
90  public void setTestCounter(int testCounter) {
91      this.testCounter = testCounter;
92  }
93
94  }
95
```

```
Source History [Icons]
1  /*
2     * To change this license header, choose License Headers in Project Properties.
3     * To change this template file, choose Tools | Templates
4     * and open the template in the editor.
5  */
6  package eregister;
7
8  import java.awt.Color;
9  import java.awt.Dimension;
10 import java.awt.event.ActionListener;
11 import java.io.File;
12 import java.io.FileNotFoundException;
13 import java.io.FileWriter;
14 import java.io.IOException;
15 import static java.lang.Long.parseLong;
16 import java.security.NoSuchAlgorithmException;
17 import java.security.SecureRandom;
18 import java.security.spec.InvalidKeySpecException;
19 import java.security.spec.KeySpec;
20 import javax.crypto.SecretKeyFactory;
21 import javax.crypto.spec.PBEKeySpec;
22 import java.util.Scanner;
23 import java.util.logging.Level;
24 import java.util.logging.Logger;
25 import javax.swing.DefaultComboBoxModel;
26 import javax.swing.JButton;
27 import javax.swing.JFrame;
28 import javax.swing.JOptionPane;
29 import javax.swing.JPanel;
30
31 /**
32  *
33  * @author Piotr
34  */
35 public class Main {
36
37     /**
38      *
39      public static String[] subjects;
40      public static Group[] groups = new Group[20];
41      public static int groupCounter;
42      public static Test[] tests = new Test[100];
43      public static int testCounter;
44      public static int chosenTest = -1;
45      public static Student chosenStudent;
46      public static boolean csFlag = false;
47      public static long day;
48      public static int period;
49      public static String[] gNames;
50      public static String[] tNames;
51      public static String temp1 = "default";
52      public static String login = "admin";
```

```
52 public static String login = "admin";
53 public static String password = "password";
54 public static int num1;
55 public static int num2;
56 public static int var = -1;
57 public static boolean value = false;
58 public static int chosenGroup = -1;
59 public static byte[] lSalt = new byte[16];
60 public static byte[] pSalt = new byte[16];
61
62 public static JFrame frame = new JFrame();
63 public static Panell window3 = new Panell(); //calendar edit
64 public static Panell window5 = new Panell(); //marks add marks for a test
65 public static Panell window6 = new Panell(); //marks view/edit all marks
66 public static Panell window7 = new Panell(); //marks add/edit
67 public static Panel2 window8 = new Panel2(); //main menu
68 public static Panelll window9 = new Panelll(); //marks panel menu
69 public static Panel3 window16 = new Panel3(); //login panel
70 public static Panel4 window17 = new Panel4(); //marks 5-student panel
71 public static Grid window27 = new Grid(true, false, false); //attendance panel
72 public static Grid window28 = new Grid(false, false, true); //calendar panel
73 public static Grid window29 = new Grid(false, true, false); //marks grid
74 public static Panel3 window31 = new Panel3(); //attendance student choice panel
75
76 public static byte[] encodeText(String text, int i) throws InvalidKeySpecException, NoSuchAlgorithmException {
77     SecureRandom random = new SecureRandom();
78     byte[] salt = new byte[16];
79     random.nextBytes(salt);
80     switch (i) {
81         case 0: //we save actual login
82             lSalt = salt;
83             break;
84         case 1: //we save actual password
85             pSalt = salt;
86             break;
87         case 2: //we save inputted login
88             salt = lSalt;
89             break;
90         case 3: //we save inputted password
91             salt = pSalt;
92             break;
93         default:
94             break;
95     }
96
97     KeySpec spec = new PBEKeySpec(text.toCharArray(), salt, 65536, 128);
98     SecretKeyFactory factory = SecretKeyFactory.getInstance("PBKDF2WithHmacSHA1");
99
100     byte[] hash = factory.generateSecret(spec).getEncoded();
101     return hash;
102 }
103
```

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103
104 public static void savePassword(String password, String s, int i) throws InvalidKeySpecException, NoSuchAlgorithmException {
105     writeToFile(s, new String(encodeText(password, i)));
106 }
107
108 public static String readPassword(String s) {
109     String a = readFromFile(s);
110     return a;
111 }
112
113 //load tests in the calendar test edition screen
114 public static void listeners() {
115     ActionListener listener2 = new ActionListener() {
116         @Override
117         public void actionPerformed(java.awt.event.ActionEvent e) {
118             for (int i = 0; i < 12; i++) {
119                 for (int j = 0; j < 5; j++) {
120                     if (e.getSource() == window28.button3[1 + i][1 + j]) {
121                         chosenTest = -1;
122                         for (int k = 0; k < testCounter; k++) {
123                             if (tests[k].getName().equals(window28.button3[1 + i][1 + j].getText())) {
124                                 chosenTest = k;
125                                 window3.field1.setText(tests[chosenTest].getName());
126                                 for (int l = 0; l < 12; l++) {
127                                     if (tests[chosenTest].getSubject().equals(subjects[l])) {
128                                         window3.combol[0].setSelectedIndex(l);
129                                     }
130                                 }
131                                 for (int m = 0; m < groupCounter; m++) {
132                                     if (tests[chosenTest].getGroup().getName().equals(groups[m].getName())) {
133                                         window3.combol[1].setSelectedItem(gNames[m]);
134                                         window3.combol[1].setSelectedIndex(m);
135                                         var = m;
136                                     }
137                                 }
138                             }
139                         }
140                     if (chosenTest == -1) {
141                         window3.field1.setText("");
142                         window3.combol[0].setSelectedIndex(0);
143                         window3.combol[1].setSelectedItem(gNames[0]);
144                     }
145                 }
146             }
147         }
148     };
149
150     for (int i = 0; i < 12; i++) {
151         for (int j = 0; j < 5; j++) {
152             window28.button3[1 + i][1 + j].addActionListener(listener2);
153         }
154     }
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154     }
155 }
156 }
157
158 public static String readFromFile(String s) {
159     String output = "";
160     try {
161         File myObj = new File(s);
162         Scanner myReader = new Scanner(myObj);
163         while (myReader.hasNextLine()) {
164             String data = myReader.nextLine();
165             output = output + data;
166         }
167         myReader.close();
168     } catch (FileNotFoundException e) {
169         System.out.println("An error occurred.");
170         e.printStackTrace();
171     }
172     return output;
173 }
174
175 //attendance of the students in groups is set
176 public static void readAttendance(String s) {
177     String a = readFromFile(s);
178     String[] b = a.split(" ", 0);
179     for (int i = 0; i < b.length; i++) {
180         if ((i + 1) % 5 == 0) {
181             groups[Integer.parseInt(b[i - 4])]
182                 .getStudents()[Integer.parseInt(b[i - 3])]
183                 .setAtt(Integer.parseInt(b[i - 2]),
184                     Integer.parseInt(b[i - 1]),
185                     Integer.parseInt(b[i]));
186         }
187     }
188 }
189
190 //create array tests and fill it with objects
191 public static void readCalendar(String s, String ss) {
192     if (readFromFile("testCounter.txt").equals("") == false) {
193         testCounter = Integer.parseInt(readFromFile("testCounter.txt"));
194         tests = new Test[testCounter];
195         for (int i = 0; i < testCounter; i++) {
196             tests[i] = new Test();
197         }
198     }
199
200     String a = readFromFile(s); //unmarked tests
201     if (a.equals("") == false) {
202         String[] b = a.split(" ", 0);
203         for (int i = 0; i < b.length; i++) {
204             if ((i + 1) % 6 == 0 && i != 0) {
205                 tests[Integer.parseInt(b[i - 5])].setName(b[i - 4]);

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205         tests[Integer.parseInt(b[i - 5])].setName(b[i - 4]);
206         int p = -1;
207         for (int j = 0; j < groupCounter; j++) {
208             if (groups[j].getName().equals(b[i - 3])) {
209                 p = j;
210             }
211         }
212         if (p != -1) {
213             tests[Integer.parseInt(b[i - 5])].setGroup(groups[p]);
214         }
215         tests[Integer.parseInt(b[i - 5])].setDay(parseLong(b[i - 2]));
216         tests[Integer.parseInt(b[i - 5])].setPeriod(Integer.parseInt(b[i - 1]));
217         tests[Integer.parseInt(b[i - 5])].setSubject(b[i]);
218     }
219 }
220
221 String c = readFromFile(ss);
222 if (c.equals("") == false) { //marked tests
223     String[] d = c.split(" ", 0);
224     for (int i = 0; i < d.length; i++) {
225         String[] f = d[i].split(" ", 0);
226         tests[Integer.parseInt(f[0])].setName(f[1]);
227         int p = -1;
228         for (int n = 0; n < groupCounter; n++) {
229             if (groups[n].getName().equals(f[2])) {
230                 p = n;
231             }
232         }
233         if (p != -1) {
234             tests[Integer.parseInt(f[0])].setGroup(groups[p]);
235         }
236         tests[Integer.parseInt(f[0])].setDay(parseLong(f[3]));
237         tests[Integer.parseInt(f[0])].setPeriod(Integer.parseInt(f[4]));
238         tests[Integer.parseInt(f[0])].setSubject(f[5]);
239         int k = Integer.parseInt(f[6]);
240         Mark[] mhm = new Mark[k];
241         if (p != -1) {
242             for (int r = 0; r < k; r++) {
243                 mhm[r] = new Mark(groups[p].getStudents()[r],
244                 tests[Integer.parseInt(f[0])], Integer.parseInt(f[r + 7]));
245             }
246         }
247         tests[Integer.parseInt(f[0])].setMarks(mhm);
248         tests[Integer.parseInt(f[0])].marked = true;
249     }
250 }
251 }
252 }
253
254 //create array groups and fill it
255 public static void readGroups(String ss) {
256     if (readFromFile("groupCounter.txt").equals("") == false) {

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256     if (readFromFile("groupCounter.txt").equals("") == false) {
257         groupCounter = Integer.parseInt(readFromFile("groupCounter.txt"));
258         groups = new Group[groupCounter];
259         for (int i = 0; i < groupCounter; i++) {
260             groups[i] = new Group();
261         }
262     }
263
264     String c = readFromFile(ss);
265     if (c.equals("") == false) {
266         String[] d = c.split(" ", 0);
267         for (int i = 0; i < d.length; i++) {
268             String[] f = d[i].split(" ", 0);
269             groups[i].setName(f[0]);
270             groups[i].setCardinality(Integer.parseInt(f[1]));
271             Student[] chads = new Student(Integer.parseInt(f[1]));
272             for (int j = 0; j < Integer.parseInt(f[1]); j++) {
273                 chads[j] = new Student(f[2 + j]);
274             }
275             groups[i].setStudents(chads);
276         }
277     }
278 }
279
280 public static void writeToFile(String s, String t) {
281     try {
282         File myObj = new File(s);
283         if (myObj.createNewFile()) {
284             } else {
285             }
286     } catch (IOException e) {
287         System.out.println("An error occurred.");
288         e.printStackTrace();
289     }
290     try {
291         FileWriter myWriter = new FileWriter(s);
292         myWriter.flush();
293         myWriter.write(t);
294         myWriter.close();
295     } catch (IOException e) {
296         System.out.println("An error occurred.");
297         e.printStackTrace();
298     }
299 }
300
301 // save integers describing attendance of students in student array with spaces as stopwords
302 public static void saveAttendance() {
303     String text = "";
304     for (int i = 0; i < groupCounter; i++) {
305         for (int j = 0; j < groups[i].getCardinality(); j++) {
306             for (int k = 0; k < 12; k++) {
307                 for (int m = 0; m < 50; m++) {

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307         for (int m = 0; m < 50; m++) {
308             text = text + Integer.toString(i) + " "
309                     + Integer.toString(j) + " "
310                     + Integer.toString(k) + " "
311                     + Integer.toString(m)
312                     + " " + Integer.toString(groups[i].getStudents()[j].getAtt(k, m)) + " ";
313         }
314     }
315 }
316 }
317 writeToFile("attendance.txt", text);
318
319 }
320
321 public static void saveCalendar() {
322     String text = "";
323     String text2 = "";
324     for (int i = 0; i < testCounter; i++) {
325         String storr = "";
326         if (tests[i].marked == true) {
327             for (int j = 0; j < tests[i].getMarks().length; j++) {
328                 storr = storr + tests[i].getMarks()[j].getNumber() + " ";
329             }
330             text2 = text2 + Integer.toString(i)
331                     + " " + tests[i].getName()
332                     + " " + tests[i].getGroup().getName()
333                     + " " + Long.toString(tests[i].getDay())
334                     + " " + Integer.toString(tests[i].getPeriod())
335                     + " " + tests[i].getSubject()
336                     + " " + tests[i].getMarks().length
337                     + " " + storr + " ";
338         } else {
339             text = text + Integer.toString(i)
340                     + " " + tests[i].getName()
341                     + " " + tests[i].getGroup().getName()
342                     + " " + Long.toString(tests[i].getDay())
343                     + " " + Integer.toString(tests[i].getPeriod())
344                     + " " + tests[i].getSubject() + " ";
345         }
346     }
347     writeToFile("calendar.txt", text);
348     writeToFile("testCounter.txt", Integer.toString(testCounter));
349     writeToFile("calendarMarked.txt", text2);
350 }
351
352 public static void saveGroups() {
353     String text = "";
354     for (int i = 0; i < groupCounter; i++) {
355         text = text + groups[i].getName() + " " + groups[i].getCardinality() + " ";
356         for (int j = 0; j < groups[i].getCardinality(); j++) {
357             text = text + groups[i].getStudents()[j].getName() + " ";
358         }

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Source History
358         }
359         text = text + "    ";
360     }
361     writeToFile("groupCounter.txt", Integer.toString(groupCounter));
362     writeToFile("groups.txt", text);
363 }
364
365 public static void initFrame() {
366     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
367     frame.setVisible(true);
368     frame.setSize(1000, 1000);
369 }
370
371 public static void addPanel(JPanel panel) {
372     frame.add(panel);
373     frame.pack();
374     frame.setLocationRelativeTo(null);
375 }
376
377 public static void dispTests() {
378     String str = "";
379     for (int i = 0; i < testCounter; i++) {
380         str = str + tests[i].toString() + " ";
381         for (int j = 0; j < tests[i].getGroup().getCardinality(); j++) {
382             if (tests[i].marked == true) {
383                 str = str + tests[i].getMarks()[j].toString() + " ";
384             }
385         }
386         str = str + "\n";
387     }
388     System.out.println(str);
389 }
390
391 public static void dispGroups() {
392     String str = "";
393     for (int i = 0; i < groupCounter; i++) {
394         str = str + groups[i].toString() + " ";
395         for (int j = 0; j < groups[i].getCardinality(); j++) {
396             str = str + groups[i].getAtt(j).getName() + " ";
397         }
398
399         str = str + "\n";
400     }
401     System.out.println(str);
402 }
403
404 public static void update() {
405
406     for (int i = 0; i < 12; i++) {
407         for (int j = 0; j < 5; j++) {
408             window28.button3[1 + i][1 + j].setText("");
409             boolean found = false;

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409         boolean found = false;
410         int num = 0;
411         for (int k = 0; k < testCounter; k++) {
412             if ((tests[k].getDay() == window28.counter + j) && (tests[k].getPeriod() == i)) {
413                 found = true;
414                 num = k;
415             }
416         }
417         if (found == true) {
418             window28.button3[1 + i][1 + j].setText(tests[num].getName());
419         } else {
420             window28.button3[1 + i][1 + j].setText("");
421         }
422     }
423 }
424
425 tNames = new String[testCounter];
426
427 for (int i = 0; i < testCounter; i++) {
428     tNames[i] = tests[i].getName() + " (" + tests[i].getGroup().getName() + ")";
429 }
430
431 gNames = new String[groupCounter];
432
433 for (int i = 0; i < groupCounter; i++) {
434     gNames[i] = groups[i].getName();
435 }
436
437 window3.combol[1].setModel(new DefaultComboBoxModel(gNames));
438 window5.combol[0].setModel(new DefaultComboBoxModel(gNames));
439 window6.combol[0].setModel(new DefaultComboBoxModel(gNames));
440 listeners();
441 window5.combol[1].setModel(new DefaultComboBoxModel(tNames));
442
443 if (chosenTest != -1 && tests[chosenTest].marked == true) {
444     window3.combol[1].setSelectedIndex(var);
445     for (int j = window17.counter; j < window17.counter + 5; j++) {
446         if (j < tests[chosenTest].getMarks().length) {
447             window17.combol[j - window17.counter]
448                 .setSelectedIndex(tests[chosenTest].getMarks()[j].getNumber() - 1);
449         }
450     }
451 }
452
453 if (chosenTest != -1 && tests[chosenTest].marked == false) {
454     window3.combol[1].setSelectedIndex(var);
455     for (int j = window17.counter; j < window17.counter + 5; j++) {
456         window17.combol[j - window17.counter].setSelectedIndex(0);
457     }
458 }
459
460 if (csFlag == true) {

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460     if (csFlag == true) {
461         for (int i = 0; i < 12; i++) {
462             for (int j = 0; j < 5; j++) {
463                 if (chosenStudent.getAttendance()[i][j + window27.counter] == 1) {
464                     window27.button3[1 + i][1 + j].setBackground(Color.GREEN);
465                 } else {
466                     window27.button3[1 + i][1 + j].setBackground(Color.RED);
467                 }
468             }
469         }
470     }
471
472     for (int i = 0; i < 12; i++) {
473         for (int j = 0; j < 5; j++) {
474             window28.button3[1 + i][1 + j].setText("");
475             boolean found = false;
476             int num = 0;
477             for (int k = 0; k < Main.testCounter; k++) {
478                 if ((Main.tests[k].getDay() == window28.counter + j) && (Main.tests[k].getPeriod() == i)) {
479                     found = true;
480                     num = k;
481                 }
482             }
483             if (found == true) {
484                 window28.button3[1 + i][1 + j].setText(Main.tests[num].getName());
485             } else {
486                 window28.button3[1 + i][1 + j].setText("");
487             }
488         }
489     }
490
491     if (chosenGroup != -1) {
492         for (int i = 0; i < 5; i++) {
493             if ((window17.counter + i) < groups[chosenGroup].getCardinality()) {
494                 window17.labell[i].setText(groups[chosenGroup].getStudents()[window17.counter + i].getName());
495             }
496         }
497         for (int i = 0; i < 5; i++) {
498             if ((window17.counter + i) >= groups[chosenGroup].getCardinality()) {
499                 window17.labell[i].setText("null");
500             }
501         }
502     }
503
504     int k = 0;
505     for (int i = 0; i < groupCounter; i++) {
506         for (int j = 0; j < groups[i].getCardinality(); j++) {
507             k++;
508         }
509     }
510     String[] array1 = new String[k];
511     k = 0;

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511     k = 0;
512     for (int i = 0; i < groupCounter; i++) {
513         for (int j = 0; j < groups[i].getCardinality(); j++) {
514             array1[k] = groups[i].getStudents()[j].getName();
515             k++;
516         }
517     }
518     window31.combol.setModel(new DefaultComboBoxModel(array1));
519     frame.repaint();
520
521 }
522
523 public static void changePanel(JButton button, JPanel panell, JPanel panel2) {
524     ActionListener listener = new ActionListener() {
525         @Override
526         public void actionPerformed(java.awt.event.ActionEvent e) {
527             if (e.getSource() == button && e.getSource() != window16.button1[1]
528                 && e.getSource() != window3.button1[2] && e.getSource() != window5.button1[2]
529                 && e.getSource() != window17.button3[1] && e.getSource() != window5.button1[1]
530                 && e.getSource() != window6.button1[2] && e.getSource() != window6.button1[1]
531                 && e.getSource() != window8.button1[5] && e.getSource() != window31.button1[1]
532                 && e.getSource() != window8.button1[7] && e.getSource() != window29.button2) {
533                 frame.remove(panell);
534                 frame.add(panel2);
535                 frame.pack();
536                 frame.setLocationRelativeTo(null);
537                 update();
538                 frame.repaint();
539             }
540             if (e.getSource() == window29.button2) {
541                 window29.counter = 0;
542                 frame.remove(panell);
543                 frame.add(panel2);
544                 frame.pack();
545                 frame.setLocationRelativeTo(null);
546                 update();
547                 frame.repaint();
548             }
549             if (e.getSource() == window8.button1[7]) {
550                 saveGroups();
551                 saveAttendance();
552                 saveCalendar();
553                 frame.remove(panell);
554                 frame.add(panel2);
555                 frame.pack();
556                 frame.setLocationRelativeTo(null);
557                 update();
558                 frame.repaint();
559             }
560             if (e.getSource() == window8.button1[5]) {
561                 String name = JOptionPane.showInputDialog(frame, "What's the name of the group?");
562                 frame.repaint();

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Source History
562     frame.repaint();
563     String count = JOptionPane.showInputDialog(frame, "How many students does it have?");
564     frame.repaint();
565     try {
566         String[] pal = new String[Integer.parseInt(count)];
567         Student[] xps = new Student[Integer.parseInt(count)];
568         boolean variabel = true;
569         for (int i = 0; i < Integer.parseInt(count); i++) {
570             pal[i] = JOptionPane.showInputDialog(frame, "What's the name of the next student?");
571             frame.repaint();
572         }
573         for (int i = 0; i < Integer.parseInt(count); i++) {
574             xps[i] = new Student(pal[i]);
575             if (pal[i].equals("")) {
576                 variabel = false;
577             }
578         }
579         if (name.equals("") == false && Integer.parseInt(count) > 0 && variabel == true) {
580             Group[] temp = new Group[groupCounter];
581             for (int i = 0; i < groupCounter; i++) {
582                 temp[i] = groups[i];
583             }
584             groups = new Group[groupCounter + 1];
585             for (int i = 0; i < groupCounter; i++) {
586                 groups[i] = temp[i];
587             }
588             groups[groupCounter] = new Group(name, Integer.parseInt(count));
589             groups[groupCounter].setStudents(xps);
590             groupCounter++;
591         }
592     } catch (Exception exc) {
593         System.out.println(exc);
594     }
595
596     update();
597 }
598 if (e.getSource() == window6.button1[1]) {
599     chosenGroup = -1;
600     frame.remove(panel1);
601     frame.add(panel2);
602     frame.pack();
603     frame.setLocationRelativeTo(null);
604     update();
605     frame.repaint();
606 }
607 if (e.getSource() == window6.button1[2]) {
608     chosenGroup = window6.combol[0].getSelectedIndex();
609     window29.label1[0].setText(groups[chosenGroup].getName());
610
611     Student[] ttt = groups[chosenGroup].getStudents();
612     String[] sss = new String[groups[chosenGroup].getCardinality()];
613     for (int j = 0; j < groups[chosenGroup].getCardinality(); j++) {

```


Group.java × Main.java × Mark.java × Student.java × Test.java × Grid.java × Panel3.java × Panel4.java ×

Source History

```
613         for (int j = 0; j < groups[chosenGroup].getCardinality(); j++) {
614             sss[j] = ttt[j].getName();
615         }
616         for (int i = 0; i < 5; i++) {
617             if ((window29.counter + i) < groups[chosenGroup].getCardinality()) {
618                 window29.button3[0][1 + i].setText(sss[window29.counter + i]);
619             } else {
620                 window29.button3[0][1 + i].setText("");
621             }
622         }
623         String[][] m = new String[12][groups[chosenGroup].getCardinality()];
624         for (int i = 0; i < 12; i++) {
625             for (int j = 0; j < groups[chosenGroup].getCardinality(); j++) {
626                 m[i][j] = "";
627             }
628         }
629         for (int i = 0; i < testCounter; i++) {
630             if (tests[i].getGroup().getName().equals(groups[chosenGroup].getName())) {
631                 for (int j = 0; j < groups[chosenGroup].getCardinality(); j++) {
632                     int subb = -1;
633                     for (int k = 0; k < subjects.length; k++) {
634                         if (tests[i].getSubject().equals(subjects[k])) {
635                             subb = k;
636                         }
637                     }
638                     if (tests[i].marked == true) {
639                         m[subb][j] = m[subb][j] + tests[i].getMarks()[j].getNumber() + ", ";
640                     }
641                 }
642             }
643         }
644         for (int i = 0; i < 12; i++) {
645             for (int j = 0; j < 5; j++) {
646                 if (j < groups[chosenGroup].getCardinality() && m[i][j].equals("") == false) {
647                     window29.button3[1 + i][1 + window29.counter + j].setText(m[i][j]);
648                 } else {
649                     window29.button3[1 + i][1 + window29.counter + j].setText("");
650                 }
651             }
652         }
653
654         frame.remove(panell1);
655         frame.add(panel2);
656         frame.pack();
657         frame.setLocationRelativeTo(null);
658         update();
659         frame.repaint();
660     }
661     if (e.getSource() == window17.button3[1]) {
662         for (int i = 0; i < 5; i++) {
663             window17.arroy[window17.counter + i] = window17.combol[i].getSelectedIndex() + 1;
664         }
```



```

715         int j = 0;
716         for (int i = 0; i < testCounter; i++) {
717             temp[i] = tests[i];
718         }
719         tests = new Test[testCounter - 1];
720         for (int i = 0; i < testCounter; i++) {
721             if (i != chosenTest) {
722                 tests[j] = temp[i];
723                 j++;
724             }
725         }
726         testCounter--;
727         chosenGroup = -1;
728         chosenTest = -1;
729         update();
730
731     }
732     if (e.getSource() == window3.button1[1]) {
733         chosenGroup = -1;
734         chosenTest = -1;
735     }
736     if (e.getSource() == window3.button1[2]) {
737         boolean value2 = true;
738         int chosenGroup = -1;
739         chosenGroup = window3.combol[1].getSelectedIndex();
740
741         for (int i = 0; i < testCounter; i++) {
742             if (window3.field1.getText().equals(tests[i].getName())) {
743                 value2 = false;
744             }
745         }
746
747         if (chosenTest == -1) {
748             if (value2 == true) {
749                 Test[] temp = new Test[testCounter];
750                 int j = 0;
751                 for (int i = 0; i < testCounter; i++) {
752                     temp[i] = tests[i];
753                 }
754                 tests = new Test[testCounter + 1];
755                 for (int i = 0; i < testCounter; i++) {
756                     tests[i] = temp[i];
757                 }
758
759                 tests[testCounter] = new Test(window3.field1.getText(),
760                     groups[chosenGroup], window28.counter + num2, num1,
761                     subjects[window3.combol[0].getSelectedIndex()]);
762                 testCounter++;
763                 frame.remove(panel1);
764                 frame.add(panel2);
765                 frame.pack();
766                 frame.setLocationRelativeTo(null);

```

```

766         frame.setLocationRelativeTo(null);
767         update();
768     }
769 } else {
770
771     tests[chosenTest].setName(window3.field1.getText());
772     tests[chosenTest].setSubject(subjects[window3.combol[0].getSelectedIndex()]);
773     tests[chosenTest].setGroup(groups[chosenGroup]);
774     frame.remove(panel1);
775     frame.add(panel2);
776     frame.pack();
777     frame.setLocationRelativeTo(null);
778     update();
779 }
780 chosenGroup = -1;
781 chosenTest = -1;
782 }
783 if (e.getSource() == window16.button1[1] &&
784     window16.field1[0].getText().equals("") == false
785     && window16.field1[1].getText().equals("") == false) {
786     try {
787         savePassword(window16.field1[0].getText(), "login2.txt", 2);
788         savePassword(window16.field1[1].getText(), "password2.txt", 3);
789     } catch (InvalidKeySpecException ex) {
790         Logger.getLogger(Main.class.getName()).log(Level.SEVERE, null, ex);
791     } catch (NoSuchAlgorithmException ex) {
792         Logger.getLogger(Main.class.getName()).log(Level.SEVERE, null, ex);
793     }
794     if (readPassword("login.txt").equals(readPassword("login2.txt"))
795         && readPassword("password.txt").equals(readPassword("password2.txt"))) {
796         readGroups("groups.txt");
797         readAttendance("attendance.txt");
798         readCalendar("calendar.txt", "calendarMarked.txt");
799         window16.field1[0].setText("");
800         window16.field1[1].setText("");
801         frame.remove(panel1);
802         frame.add(panel2);
803         frame.pack();
804         frame.setLocationRelativeTo(null);
805         update();
806     }
807 }
808 for (int i = 0; i < 12; i++) {
809     for (int j = 0; j < 5; j++) {
810         if (e.getSource() == window28.button3[1 + i][1 + j]) {
811             num1 = i;
812             num2 = j;
813         }
814     }
815 }
816
817 if (e.getSource() == window31.button1[1]) {

```

Group.java x Main.java x Mark.java x Student.java x Test.java x Grid.java x Panel3.java x Pan

Source History

```
817         if (e.getSource() == window31.button1[1]) {
818             int k = 0;
819             for (int i = 0; i < groupCounter; i++) {
820                 for (int j = 0; j < groups[i].getCardinality(); j++) {
821                     if (k == window31.combol.getSelectedIndex()) {
822                         chosenStudent = groups[i].getStudents()[j];
823                     }
824                     k++;
825                 }
826             }
827             window27.labell[0].setText(chosenStudent.getName());
828             for (int i = 0; i < 12; i++) {
829                 for (int j = 0; j < 5; j++) {
830                     if (chosenStudent.getAttendance()[i][j + window27.counter] == 1) {
831                         window27.button3[1 + i][1 + j].setBackground(Color.GREEN);
832                     } else {
833                         window27.button3[1 + i][1 + j].setBackground(Color.RED);
834                     }
835                 }
836             }
837             frame.remove(panell1);
838             frame.add(panel2);
839             frame.pack();
840             frame.setLocationRelativeTo(null);
841             update();
842             frame.repaint();
843         }
844
845         if (e.getSource() == window27.button2) {
846             window27.counter = 0;
847             csFlag = false;
848             window27.labell[1].setText("week " + ((window27.counter / 5) + 1));
849             window27.labell[0].setText("student");
850         }
851     }
852 ;
853 };
854 button.addActionListener(listener);
855
856 }
857
858 public static void exit(JButton button) {
859     ActionListener listener = new ActionListener() {
860         @Override
861         public void actionPerformed(java.awt.event.ActionEvent e) {
862             if (e.getSource() == button) {
863                 System.exit(0);
864             }
865         }
866     };
867 };
868 button.addActionListener(listener);
```

```
868         button.addActionListener(listener);
869     }
870
871     public static void initWindows() {
872
873         // Panell
874         window3.label[2].setText("group:");
875         window3.label2.setText("type group name:");
876         window3.combol[0].setModel(new DefaultComboBoxModel(subjects));
877
878         window5.remove(window5.button1[0]);
879         window5.remove(window5.label[0]);
880         window5.remove(window5.field1);
881         window5.label[1].setText("group:");
882         window5.label[2].setText("test:");
883         window5.remove(window5.label2);
884         window5.remove(window5.field2);
885         window5.label[1].setPreferredSize(new Dimension(70, 30));
886         window5.label[2].setPreferredSize(new Dimension(70, 30));
887         window5.remove(window5.label[1]);
888         window5.remove(window5.combol[0]);
889
890         window6.remove(window6.button1[0]);
891         window6.remove(window6.label[0]);
892         window6.remove(window6.field1);
893         window6.remove(window6.label[2]);
894         window6.remove(window6.combol[1]);
895         window6.label[1].setText("group:");
896         window6.remove(window6.label2);
897         window6.remove(window6.field2);
898
899         window7.remove(window7.button1[0]);
900         window7.remove(window7.label[0]);
901         window7.remove(window7.field1);
902         window7.label[1].setText("test: ");
903         window7.remove(window7.label2);
904         window7.remove(window7.field2);
905
906         //Panel2
907         window8.button1[7].setText("logout and save");
908         window8.button1[5].setText("add group");
909         window8.remove(window8.button1[3]);
910         window8.remove(window8.button1[4]);
911         window8.remove(window8.button1[6]);
912
913         window16.remove(window16.label1[0]);
914         window16.remove(window16.combol);
915         window16.remove(window16.label1[3]);
916         window16.remove(window16.field1[2]);
917         window16.button1[0].setText("exit");
918
919         window31.label1[0].setText("student:");
```

Group.java × Main.java × Mark.java × Student.java × Test.java × Grid.java × Panel3.java ×

Source History

```
919 window31.labell[0].setText("student:");
920 window31.remove(window31.labell[1]);
921 window31.remove(window31.labell[2]);
922 window31.remove(window31.labell[3]);
923 window31.remove(window31.fieldl[0]);
924 window31.remove(window31.fieldl[1]);
925 window31.remove(window31.fieldl[2]);
926 int k = 0;
927 for (int i = 0; i < groupCounter; i++) {
928     for (int j = 0; j < groups[i].getCardinality(); j++) {
929         k++;
930     }
931 }
932 String[] arrayl = new String[k];
933 k = 0;
934 for (int i = 0; i < groupCounter; i++) {
935     for (int j = 0; j < groups[i].getCardinality(); j++) {
936         arrayl[k] = groups[i].getStudents()[j].getName();
937         k++;
938     }
939 }
940 window31.combol.setModel(new DefaultComboBoxModel(arrayl));
941 frame.repaint();
942
943 //Panel9 is initialized in the declaration at the beginning of this class
944 //Grid
945 window28.remove(window28.labell[0]);
946 window28.label12.setText("calendar");
947
948 for (int i = 0; i < 12; i++) {
949     window29.button3[1 + i][0].setText("subject " + i);
950 }
951 for (int i = 0; i < 5; i++) {
952     window29.button3[0][1 + i].setText("student " + i);
953 }
954 window29.labell[0].setText("group name");
955 window29.remove(window29.labell[1]);
956 window29.label12.setText("marks");
957 for (int i = 0; i < 12; i++) {
958     window29.button3[1 + i][0].setText(subjects[i]);
959 }
960 }
961
962 public static void initData() throws InvalidKeySpecException, NoSuchAlgorithmException {
963     savePassword("admin", "login.txt", 0);
964     savePassword("password", "password.txt", 1);
965     window29.counter = 0;
966     subjects = new String[]{"English", "Mathematics", "Science", "PE",
967         "History", "Computer Science", "Art", "Language A",
968         "Subject 9", "Subject 10", "Subject 11", "Subject 12"};
969     gNames = new String[groupCounter];
970 }
```

```

970
971     for (int i = 0; i < groupCounter; i++) {
972         gNames[i] = groups[i].getName();
973     }
974
975     tNames = new String[testCounter];
976
977     for (int i = 0; i < testCounter; i++) {
978         tNames[i] = tests[i].getName();
979     }
980 }
981
982 public static void logic() {
983
984     //login panel
985     addPanel(window16);
986     exit(window16.button1[0]);
987     changePanel(window16.button1[1], window16, window8);
988
989     //main menu
990     changePanel(window8.button1[0], window8, window31);
991     changePanel(window8.button1[1], window8, window28);
992     changePanel(window8.button1[2], window8, window9);
993     changePanel(window8.button1[7], window8, window16);
994     changePanel(window8.button1[5], window8, window8);
995
996     //attendance panel
997     changePanel(window27.button2, window27, window8);
998     changePanel(window31.button1[0], window31, window8);
999     changePanel(window31.button1[1], window31, window27);
1000
1001     //calendar panel
1002     changePanel(window28.button2, window28, window8);
1003     for (int i = 0; i < 12; i++) {
1004         for (int j = 0; j < 5; j++) {
1005             changePanel(window28.button3[1 + i][1 + j], window28, window3);
1006         }
1007     }
1008     changePanel(window3.button1[0], window3, window28);
1009     changePanel(window3.button1[1], window3, window28);
1010     changePanel(window3.button1[2], window3, window28);
1011
1012     //marks panel
1013     changePanel(window9.button1[0], window9, window5);
1014     changePanel(window9.button1[1], window9, window6); //window6
1015     changePanel(window9.button1[2], window9, window8);
1016     //changePanel(window34.button1, window34, window8);
1017
1018     changePanel(window5.button1[1], window5, window9);
1019     changePanel(window5.button1[2], window5, window17);
1020
1021     changePanel(window6.button1[1], window6, window9);

```

```

1011
1012 //marks panel
1013 changePanel(window9.button1[0], window9, window5);
1014 changePanel(window9.button1[1], window9, window6); //window6
1015 changePanel(window9.button1[2], window9, window8);
1016 //changePanel(window34.button1, window34, window8);
1017
1018 changePanel(window5.button1[1], window5, window9);
1019 changePanel(window5.button1[2], window5, window17);
1020
1021 changePanel(window6.button1[1], window6, window9);
1022 changePanel(window6.button1[2], window6, window29);
1023
1024 changePanel(window17.button3[0], window17, window5);
1025 changePanel(window17.button3[1], window17, window5);
1026
1027 changePanel(window29.button2, window29, window6);
1028
1029 changePanel(window7.button1[1], window7, window29);
1030 changePanel(window7.button1[2], window7, window29);
1031 }
1032
1033 public static void main(String[] args) throws InvalidKeySpecException, NoSuchAlgorithmException {
1034 // TODO code application logic here
1035
1036 initData();
1037 initFrame();
1038 initWindows();
1039
1040 logic();
1041
1042 }
1043
1044 }
1045

```



```
1  ...5 lines
6  package eregister;
7
8  /**...4 lines */
12 public class Mark {
13
14     private Student student;
15     private Test test;
16     private int number;
17
18     public Mark(Student student, Test test, int number) {
19         this.student = student;
20         this.test = test;
21         this.number = number;
22     }
23
24     public Mark() {
25
26     }
27
28     @Override
29     public String toString() {
30         return "Mark{" + "student=" + student + ", test=" + test + ", number=" + number + '}';
31     }
32
33     public Student getStudent() {
34         return student;
35     }
36
37     public void setStudent(Student student) {
38         this.student = student;
39     }
40
41     public Test getTest() {
42         return test;
43     }
44
45     public void setTest(Test test) {
46         this.test = test;
47     }
48
49     public int getNumber() {
50         return number;
51     }
52
53     public void setNumber(int number) {
54         this.number = number;
55     }
56
57 }
58
```

```

1  ...5 lines
6  package eregister;
7
8  /**...4 lines */
12 public class Student {
13
14     private String name;
15     private Group group;
16     private int[][] attendance = new int[12][50];
17
18     public Student(Group group, String name) {
19         this.name = name;
20         this.group = group;
21         for (int i = 0; i < 12; i++) {
22             for (int j = 0; j < 50; j++) {
23                 attendance[i][j] = 1;
24             }
25         }
26     }
27
28     public Student() {
29         for (int i = 0; i < 12; i++) {
30             for (int j = 0; j < 50; j++) {
31                 attendance[i][j] = 1;
32             }
33         }
34     }
35
36     public Student(String name) {
37         for (int i = 0; i < 12; i++) {
38             for (int j = 0; j < 50; j++) {
39                 attendance[i][j] = 1;
40             }
41         }
42         this.name = name;
43     }
44
45     @Override
46     public String toString() {
47         return "Student{" + name + '}';
48     }
49
50     public Group getGroup() {
51         return group;
52     }
53
54     public void setGroup(Group group) {
55         this.group = group;
56     }
57
58     public int[][] getAttendance() {
59         return attendance;

```

```

51         return group;
52     }
53
54     public void setGroup(Group group) {
55         this.group = group;
56     }
57
58     public int[][] getAttendance() {
59         return attendance;
60     }
61
62     public int getAtt(int i, int j) {
63         return attendance[i][j];
64     }
65
66     public void setAttendance(int[][] attendance) {
67         this.attendance = attendance;
68     }
69
70     public void setAtt(int i, int j, int k) {
71         attendance[i][j] = k;
72     }
73
74     public String getName() {
75         return name;
76     }
77
78     public void setName(String name) {
79         this.name = name;
80     }
81
82
83
84 }
85

```

```

1   /*
2   * To change this license header, choose License Headers in Project Properties.
3   * To change this template file, choose Tools | Templates
4   * and open the template in the editor.
5   */
6  package eregister;
7
8   /**
9   *
10  * @author Piotr
11  */
12 public class Test {
13
14     private String name;
15     private Group group;
16     private long day;
17     private int period;
18     private Mark[] marks;
19     private String subject;
20     public boolean marked = false;
21
22     @Override
23      public String toString() {
24         return "Test{" + "name=" + name + ", group=" + group.getName()
25             + ", day=" + day + ", period="
26             + period + ", subject="
27             + subject + ", marked=" + marked + '}';
28     }
29
30      public Test(String name, Group group, long day,
31          int period, String subject) {
32         this.name = name;
33         this.group = group;
34         this.day = day;
35         this.period = period;
36         this.subject = subject;
37     }
38
39      public Test() {
40     }
41
42      public String getName() {
43         return name;
44     }
45
46      public void setName(String name) {
47         this.name = name;
48     }
49
50      public Group getGroup() {
51         return group;
52     }

```

```
Source History 
51         return group;
52     }
53
54     public void setGroup(Group group) {
55         this.group = group;
56     }
57
58     public long getDay() {
59         return day;
60     }
61
62     public void setDay(long day) {
63         this.day = day;
64     }
65
66     public int getPeriod() {
67         return period;
68     }
69
70     public void setPeriod(int period) {
71         this.period = period;
72     }
73
74     public Mark[] getMarks() {
75         return marks;
76     }
77
78     public void setMarks(Mark[] marks) {
79         this.marks = marks;
80     }
81
82     public String getSubject() {
83         return subject;
84     }
85
86     public void setSubject(String subject) {
87         this.subject = subject;
88     }
89
90 }
91
```

```

1  /*
2   * To change this license header, choose License Headers in Project Properties.
3   * To change this template file, choose Tools | Templates
4   * and open the template in the editor.
5   */
6   package eregister;
7
8   import static eregister.Main.chosenGroup;
9   import static eregister.Main.groups;
10  import static eregister.Main.subjects;
11  import static eregister.Main.testCounter;
12  import static eregister.Main.tests;
13  import java.awt.Color;
14  import java.awt.Dimension;
15  import java.awt.FlowLayout;
16  import java.awt.GridLayout;
17  import java.awt.event.ActionListener;
18  import javax.swing.JButton;
19  import javax.swing.JLabel;
20  import javax.swing.JPanel;
21
22  /**
23   *
24   * @author Piotr
25   *
26   */
27  // attendance panel
28  // calendar panel
29  // marks panel
30  public class Grid extends JPanel {
31
32      public JLabel label1[] = new JLabel[2];
33      public JLabel label2 = new JLabel("attendance");
34      public JButton[] button1 = new JButton[2];
35      public JButton button2 = new JButton("return");
36      public JPanel panell = new JPanel(new GridLayout(13, 6));
37      public JButton[][] button3 = new JButton[13][6];
38      public int counter = 0;
39      public String b;
40
41      public void listeners(boolean isAttendance, boolean isCalendar) {
42          ActionListener listener2 = new ActionListener() {
43              @Override
44              public void actionPerformed(java.awt.event.ActionEvent e) {
45                  for (int i = 0; i < 12; i++) {
46                      for (int j = 0; j < 5; j++) {
47                          if (e.getSource() == button3[1 + i][1 + j]) {
48                              Main.temp1 = button3[1 + i][1 + j].getText();
49                              b = button3[1 + i][1 + j].getText();
50                              if (isAttendance == true) {
51                                  if (Main.chosenStudent.getAtt(i, j + counter) == 1) {
52                                      Main.chosenStudent.setAtt(i, j + counter, 0);

```



```

52         Main.chosenStudent.setAtt(i, j + counter, 0);
53         button3[1 + i][1 + j].setBackground(Color.RED);
54     } else {
55         Main.chosenStudent.setAtt(i, j + counter, 1);
56         button3[1 + i][1 + j].setBackground(Color.GREEN);
57     }
58     } else if (isCalendar == true) {
59         Main.period = i;
60         Main.day = counter + j;
61     }
62 }
63 }
64 }
65 }
66 ;
67 };
68 for (int i = 0; i < 12; i++) {
69     for (int j = 0; j < 5; j++) {
70         button3[1 + i][1 + j].addActionListener(listener2);
71     }
72 }
73 }
74
75 public void changeComponents(Boolean isMarks, Boolean isAttendance, Boolean isCalendar) {
76     ActionListener listener = new ActionListener() {
77         @Override
78         public void actionPerformed(java.awt.event.ActionEvent e) {
79             if (e.getSource() == button1[0] && counter >= 5) {
80                 counter = counter - 5;
81             } else if (e.getSource() == button1[1] && isMarks == true
82                 && (counter + 5) <= Main.groups[chosenGroup].getCardinality() - 1) {
83                 counter = counter + 5;
84             } else if (e.getSource() == button1[1] && isMarks == false && counter <= 40) {
85                 counter = counter + 5;
86             }
87             if (isMarks == true) {
88                 Student[] ttt = Main.groups[chosenGroup].getStudents();
89                 String[] sss = new String[Main.groups[chosenGroup].getCardinality()];
90                 for (int j = 0; j < Main.groups[chosenGroup].getCardinality(); j++) {
91                     sss[j] = ttt[j].getName();
92                 }
93                 for (int i = 0; i < 5; i++) {
94                     if ((counter + i) < Main.groups[chosenGroup].getCardinality()) {
95                         button3[0][1 + i].setText(sss[counter + i]);
96                     } else {
97                         button3[0][1 + i].setText("");
98                     }
99                 }
100                 String[][] m = new String[12][Main.groups[chosenGroup].getCardinality()];
101                 for (int i = 0; i < 12; i++) {
102                     for (int j = 0; j < Main.groups[chosenGroup].getCardinality(); j++) {
103                         m[i][j] = "";

```


Group.javaMain.javaMark.javaStudent.javaTest.javaGrid.javaPanel3.javaPanel4.java

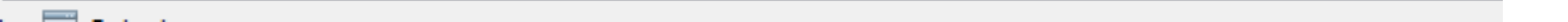
SourceHistory

```
103         m[i][j] = "";
104     }
105 }
106 for (int i = 0; i < testCounter; i++) {
107     if (tests[i].getGroup().getName().equals(groups[chosenGroup].getName())) {
108         int subb = -1;
109         for (int k = 0; k < subjects.length; k++) {
110             if (tests[i].getSubject().equals(subjects[k])) {
111                 subb = k;
112             }
113         }
114         for (int j = 0; j < groups[chosenGroup].getCardinality(); j++) {
115             if (tests[i].marked == true) {
116                 m[subb][j] = m[subb][j] + tests[i].getMarks()[j].getNumber() + ", ";
117             }
118         }
119     }
120 }
121 for (int i = 0; i < 12; i++) {
122     for (int j = 0; j < 5; j++) {
123         if ((counter + j) < groups[chosenGroup].getCardinality()
124             && m[i][counter + j].equals("") == false) {
125             button3[1 + i][1 + j].setText(m[i][counter + j]);
126         } else {
127             button3[1 + i][1 + j].setText("");
128         }
129     }
130 }
131 } else {
132     label1[1].setText("week " + ((counter / 5) + 1));
133
134     for (int i = 0; i < 12; i++) {
135         for (int j = 0; j < 5; j++) {
136             if (isAttendance == true) {
137                 if (Main.chosenStudent.getAttendance()[i][j + counter] == 1) {
138                     button3[1 + i][1 + j].setBackground(Color.GREEN);
139                 } else {
140                     button3[1 + i][1 + j].setBackground(Color.RED);
141                 }
142             } else if (isCalendar == true) {
143                 button3[1 + i][1 + j].setText("");
144                 boolean found = false;
145                 int num = 0;
146                 for (int k = 0; k < Main.testCounter; k++) {
147                     if ((Main.tests[k].getDay() == counter + j)
148                         && (Main.tests[k].getPeriod() == i)) {
149                         found = true;
150                         num = k;
151                     }
152                 }
153                 if (found == true) {
154                     button3[1 + i][1 + j].setText(Main.tests[num].getName());
```

```

154         button3[1 + i][1 + j].setText(Main.tests[num].getName());
155     } else {
156         button3[1 + i][1 + j].setText("");
157     }
158 }
159 }
160 }
161 }
162 }
163 }
164 ;
165 };
166 button1[0].addActionListener(listener);
167 button1[1].addActionListener(listener);
168 }
169
170 public Grid(Boolean isAttendance, Boolean isMarks, Boolean isCalendar) {
171     
172     
173     setLayout(new FlowLayout(FlowLayout.CENTER, 30, 30));
174     setPreferredSize(new Dimension(660, 700));
175     label1[0] = new JLabel("student");
176     label1[1] = new JLabel("week 1");
177     label1[0].setPreferredSize(new Dimension(400, 30));
178     label1[1].setPreferredSize(new Dimension(400, 30));
179     label2.setPreferredSize(new Dimension(400, 30));
180     panel1.setPreferredSize(new Dimension(600, 390));
181     for (int i = 0; i < 13; i++) {
182         for (int j = 0; j < 6; j++) {
183             button3[i][j] = new JButton();
184             panel1.add(button3[i][j]);
185             if (isAttendance == true) {
186                 if (i >= 1 && j >= 1) {
187                     button3[i][j].setBackground(Color.GREEN);
188                     button3[i][j].setOpaque(true);
189                     button3[i][j].setText("");
190                 }
191             }
192         }
193     }
194     for (int i = 0; i < 12; i++) {
195         button3[1 + i][0].setText("period " + (i + 1));
196     }
197     button3[0][1].setText("monday");
198     button3[0][2].setText("tuesday");
199     button3[0][3].setText("wednesday");
200     button3[0][4].setText("thursday");
201     button3[0][5].setText("friday");
202     button1[0] = new JButton("←");
203     button1[1] = new JButton("→");
204     
205     
206     
207     
208     add(label2);
209     add(label1[0]);
210     add(label1[1]);
211     add(panel1);

```



```

1  ...5 lines
6  package eregister;
7
8  import java.awt.Dimension;
9  import java.awt.FlowLayout;
10 import javax.swing.JButton;
11 import javax.swing.JComboBox;
12 import javax.swing.JLabel;
13 import javax.swing.JPanel;
14 import javax.swing.JTextField;
15
16 /**...9 lines */
25 public class Panel3 extends JPanel {
26
27     public JLabel[] labell = new JLabel[4];
28     public JComboBox combol = new JComboBox();
29     public JTextField[] fieldl = new JTextField[3];
30     public JButton[] buttonl = new JButton[2];
31
32     public Panel3() {
33         
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39         
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59         
        setLayout(new FlowLayout(FlowLayout.CENTER, 30, 30));
        setPreferredSize(new Dimension(400, 350));
        labell[0] = new JLabel("name:");
        labell[1] = new JLabel("login:");
        labell[2] = new JLabel("password:");
        labell[3] = new JLabel("repeat password:");
        combol.setPreferredSize(new Dimension(200, 30));
        for (int i = 0; i < 3; i++) {
            fieldl[i] = new JTextField();
            fieldl[i].setPreferredSize(new Dimension(200, 30));
        }
        buttonl[0] = new JButton("return");
        buttonl[1] = new JButton("ok");
        add(labell[0]);
        add(combol);
        add(labell[1]);
        add(fieldl[0]);
        add(labell[2]);
        add(fieldl[1]);
        add(labell[3]);
        add(fieldl[2]);
        add(buttonl[0]);
        add(buttonl[1]);
    }
}

```

Group.javaMain.javaMark.javaStudent.javaTest.javaGrid.javaPanel3.javaPanel4.javaPanel1.java

SourceHistory

```
1  ...5 lines
6  package eregister;
7
8  import static eregister.Main.chosenGroup;
9  import static eregister.Main.groups;
10 import java.awt.Dimension;
11 import java.awt.FlowLayout;
12 import java.awt.event.ActionListener;
13 import javax.swing.DefaultComboBoxModel;
14 import javax.swing.JButton;
15 import javax.swing.JComboBox;
16 import javax.swing.JLabel;
17 import javax.swing.JPanel;
18
19 /**...6 lines */
25 // marks panel test marking panel with 5 students
26 public class Panel4 extends JPanel {
27
28     public JLabel[] label1 = new JLabel[5];
29     public JComboBox[] combol = new JComboBox[5];
30     public JButton[] button2 = new JButton[2];
31     public JButton[] button3 = new JButton[2];
32     public int counter; //index of the top student
33     public int[] array = new int[30];
34
35     public void changeComponents() {
36         ActionListener listener = new ActionListener() {
37             @Override
38             public void actionPerformed(java.awt.event.ActionEvent e) {
39                 for (int i = 0; i < 5; i++) {
40                     array[counter + i] = combol[i].getSelectedIndex() + 1;
41                     combol[i].setSelectedIndex(0);
42                 }
43                 if (chosenGroup != -1) {
44                     if (e.getSource() == button2[0] && (counter - 5) >= 0) {
45                         counter = counter - 5;
46                     } else if (e.getSource() == button2[1] && (counter + 5) <= groups[chosenGroup].getCardinality() - 1) {
47                         counter = counter + 5;
48                     }
49                     for (int i = 0; i < 5; i++) {
50                         if ((counter + i) < groups[chosenGroup].getCardinality()) {
51                             label1[i].setText(groups[chosenGroup].getStudents()[counter + i].getName());
52                             if (array[counter + i] != -1) {
53                                 combol[i].setSelectedIndex(array[counter + i] - 1);
54                             }
55                         } else {
56                             label1[i].setText("null");
57                         }
58                     }
59                 }
60             }
61         };
62     }
63 }
```

Group.java ×Main.java ×Mark.java ×Student.java ×Test.java ×Grid.java ×Panel3.java ×Panel4.java

SourceHistory

```
60 }
61 ;
62 };
63 button2[0].addActionListener(listener);
64 button2[1].addActionListener(listener);
65 }
66
67 public Panel4() {
68     for (int i = 0; i < 30; i++) {
69         array[i] = 1;
70     }
71
72     setLayout(new FlowLayout(FlowLayout.CENTER, 30, 30));
73     setPreferredSize(new Dimension(350, 400));
74     for (int i = 0; i < 5; i++) {
75         label1[i] = new JLabel("student " + i + ":");
76         label1[i].setPreferredSize(new Dimension(60, 30));
77         combol[i] = new JComboBox();
78         combol[i].setPreferredSize(new Dimension(200, 30));
79     }
80     button2[0] = new JButton("↑");
81     button2[1] = new JButton("↓");
82     button3[0] = new JButton("return");
83     button3[1] = new JButton("ok");
84     for (int i = 0; i < 5; i++) {
85         add(label1[i]);
86         add(combol[i]);
87         combol[i].setModel(new DefaultComboBoxModel(new String[]{"1", "2", "3", "4", "5", "6"}));
88     }
89     add(button2[0]);
90     add(button2[1]);
91     add(button3[0]);
92     add(button3[1]);
93     changeComponents();
94 }
95
96 }
97
```

```
1  ...5 lines
6  package eregister;
7
8  import java.awt.Dimension;
9  import java.awt.FlowLayout;
10 import javax.swing.JButton;
11 import javax.swing.JComboBox;
12 import javax.swing.JLabel;
13 import javax.swing.JPanel;
14 import javax.swing.JTextField;
15
16 /**...6 lines */
22 public class Panel1 extends JPanel {
23
24     public JLabel[] label = new JLabel[3];
25     public JTextField field1 = new JTextField();
26     public JComboBox[] combol = new JComboBox[2];
27     public JButton[] button1 = new JButton[3];
28     public JLabel label2 = new JLabel("group:");
29     public JTextField field2 = new JTextField();
30
31     public Panel1() {
32          
33                
34         
35         
36         
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```



```
1  ...5 lines
6  package eregister;
7
8  import java.awt.Dimension;
9  import java.awt.FlowLayout;
10 import javax.swing.JButton;
11 import javax.swing.JPanel;
12
13  /**...6 lines */
19 public class Panell11 extends JPanel {
20
21     public JButton[] button1 = new JButton[3];
22
23     public Panell11() {
24         setLayout(new FlowLayout(FlowLayout.CENTER, 30, 30));
25         setPreferredSize(new Dimension(400, 200));
26         button1[0] = new JButton("mark test");
27         button1[1] = new JButton("all marks");
28         button1[2] = new JButton("return");
29         for (int i = 0; i < 3; i++) {
30             add(button1[i]);
31         }
32     }
33 }
34
```

```
1  ...5 lines
6  package eregister;
7
8  import java.awt.Dimension;
9  import java.awt.FlowLayout;
10 import javax.swing.JButton;
11 import javax.swing.JPanel;
12
13 /**...10 lines */
23 public class Panel2 extends JPanel {
24
25     JButton[] button1 = new JButton[8];
26
27     public Panel2() {
28         setLayout(new FlowLayout(FlowLayout.CENTER, 30 , 30));
29         setPreferredSize(new Dimension(500, 200));
30         button1[0] = new JButton("attendance");
31         button1[1] = new JButton("calendar");
32         button1[2] = new JButton("marks");
33         button1[3] = new JButton("inbox");
34         button1[4] = new JButton("announcements");
35         button1[5] = new JButton("groups");
36         button1[6] = new JButton("users");
37         button1[7] = new JButton("logout");
38         for (int i = 0; i < 8; i++) {
39             add(button1[i]);
40         }
41     }
42
43 }
44
```